

Experiment 5: Breadth First Search (BFS)

Aim

To implement Breadth First Search (BFS) using Python.

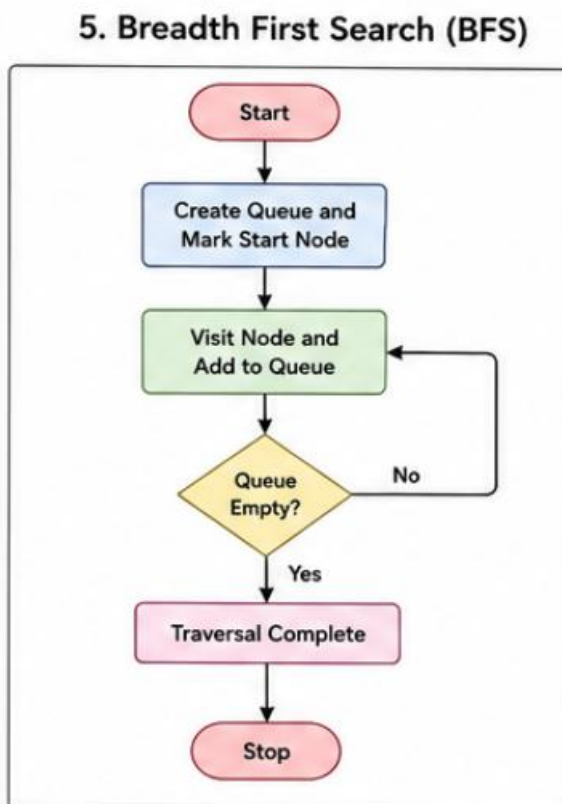
Objective

To traverse all nodes of a graph using BFS.

Algorithm

1. Create a graph.
2. Select the starting node.
3. Visit the node.
4. Add its neighbours to the queue.
5. Repeat until the queue is empty.

Flowchart

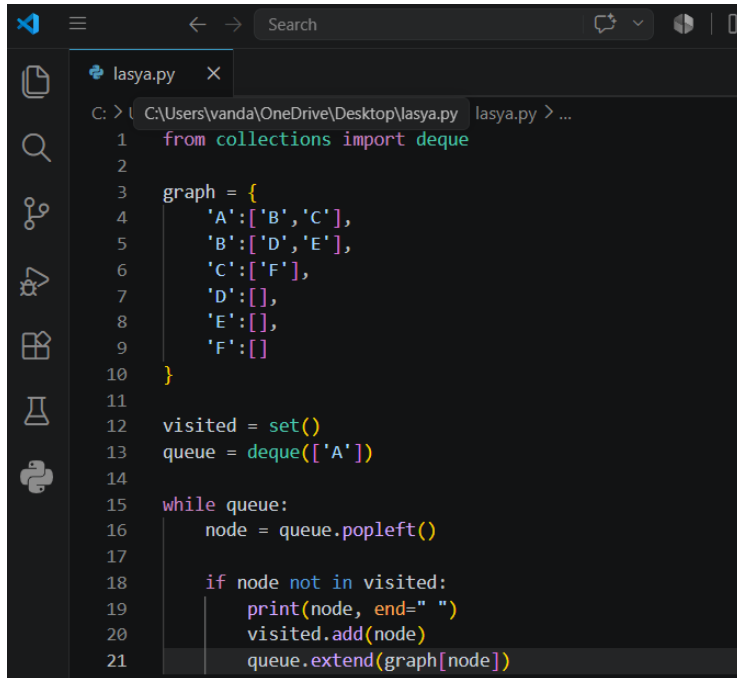


Code

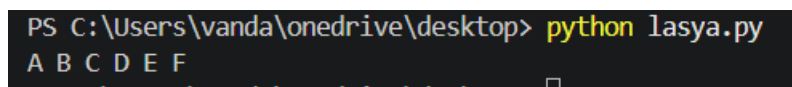
```
from collections import deque
```

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],  
    'D': [],  
    'E': [],  
    'F': []  
}  
  
visited = set()  
  
queue = deque(['A'])  
  
while queue:  
    node = queue.popleft()  
  
    if node not in visited:  
        print(node, end=" ")  
        visited.add(node)  
        queue.extend(graph[node])
```

Output :



```
1 from collections import deque
2
3 graph = {
4     'A': ['B', 'C'],
5     'B': ['D', 'E'],
6     'C': ['F'],
7     'D': [],
8     'E': [],
9     'F': []
10 }
11
12 visited = set()
13 queue = deque(['A'])
14
15 while queue:
16     node = queue.popleft()
17
18     if node not in visited:
19         print(node, end=" ")
20         visited.add(node)
21         queue.extend(graph[node])
```



```
PS C:\Users\vanda\onedrive\desktop> python lasya.py
A B C D E F
```

Result

The graph was traversed successfully using BFS.

Conclusion

The BFS algorithm visits all nodes level by level using a queue.